

AI Resume Builder - Frontend

A modern, interactive web application built with Next.js that guides users through creating professional resumes using an AI-powered (or rule-based) chatbot interface.

🚀 Features

- **Conversational UI:** A sleek, step-by-step chatbot interface to collect user information.
- **Progress Tracking:** Real-time progress bar to show how much of the resume is completed.
- **Dynamic Preview:** See your information being formatted as you provide it.
- **Instant PDF Download:** One-click generation and download of professional PDF resumes.
- **Responsive Design:** Optimized for both desktop and mobile devices.
- **Smooth Animations:** Powered by Framer Motion for a premium user experience.

🔧 Tech Stack

- **Framework:** [Next.js 15](https://nextjs.org/) (App Router)
- **Styling:** [Tailwind CSS 4](https://tailwindcss.com/)
- **UI Components:** [Radix UI](https://www.radix-ui.com/) & [Lucide Icons](https://lucide.dev/)
- **Animations:** [Framer Motion](https://www.framer.com/motion/)
- **State Management:** [Zustand](https://github.com/pmndrs/zustand)
- **API Client:** [Axios](https://axios-http.com/)
- **PDF Rendering:** [@react-pdf/renderer](https://react-pdf.org/)

📋 Prerequisites

- Node.js 18+
- npm / yarn / pnpm

⚙️ Setup & Installation

1. Clone the repository:

```
git clone <repository-url>
cd AI_RESUME_GENERATOR_FRONTEND
```

2. Install dependencies:

```
npm install
```

3. Configure Environment Variables:

Create a .env.local file in the root directory:

```
NEXT_PUBLIC_API_URL=http://localhost:8000
```

4. Run the development server:

```
npm run dev
```

5. Build for production:

```
npm run build
npm run start
```

📁 Project Structure

```
src/
├── app/                # Next.js App Router (pages and layouts)
│   ├── builder/        # Main resume builder interface
│   └── resume/         # Resume view/management
├── components/         # UI components
│   ├── chatbot/        # Chat interface components
│   ├── pdf/           # PDF generation components
│   ├── preview/       # Real-time resume preview
│   └── ui/            # Base shadcn/ui components
├── services/          # API communication (Axios)
├── store/             # Global state (Zustand)
├── types/             # TypeScript definitions
└── utils/            # Helper functions
```

🏗️ Architecture

The application follows a modern decoupled architecture with a centralized state management system.

```
graph TD
    subgraph "Frontend (Next.js)"
        UI[User Interface - React Components]
        CS[Chat Store - Zustand]
        RS[Resume Store - Zustand]
        API[API Client - Axios]
        PDF[PDF Renderer - @react-pdf]
    end

    subgraph "Backend (FastAPI)"
        BE[Resume Generation API]
    end

    UI <--> CS
    UI <--> RS
    CS --> RS
    RS --> PDF
    RS --> API
    API <--> BE
```

 Application Flow

```
sequenceDiagram
    participant User
    participant Chat as Chat Interface
    participant Store as Zustand Store
    participant Preview as Live Preview
    participant API as Backend API
    participant PDF as PDF Generator

    User->>Chat: Provides Personal Info
    Chat->>Store: Updates Resume Data
    Store->>Preview: Triggers Re-render
    Preview-->>User: Shows Live Change

    Note over User, Preview: Repeat for Experience, Skills, etc.

    User->>Chat: Completes Final Step
    Chat->>Store: Finalize Data
    Store->>API: Send Data for Backend Processing
    API-->>Store: Returns Processed Data/Blob
    Store->>PDF: Generate Downloadable PDF
    PDF-->>User: Triggers File Download
```

 Integration

This frontend is designed to work seamlessly with the [AI Resume Builder Backend \(https://github.com/Yashwant-Rangrej/AI_RESUME_GENERATOR_BACKEND\)](https://github.com/Yashwant-Rangrej/AI_RESUME_GENERATOR_BACKEND). Ensure the backend is running and the NEXT_PUBLIC_API_URL is correctly configured.

 Developed By

Yashwant Rangrej

- [GitHub \(https://github.com/Yashwant-Rangrej\)](https://github.com/Yashwant-Rangrej)
- [LinkedIn \(https://www.linkedin.com/in/yashwant-rangrej-0856993a8/\)](https://www.linkedin.com/in/yashwant-rangrej-0856993a8/)